

CATERBOT: DESIGN AND DEVELOPMENT OF A CATERPILLAR-LIKE RESCUE ROBOT FOR CONFINED AND HAZARDOUS ENVIRONMENTS

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1. Introduction

Catastrophes caused by natural and man-made disasters significantly impact human life, especially when affected individuals are trapped in confined or hazardous spaces, such as collapsed buildings, tunnels, and rubble (Lagmay et al., 2021). Search and rescue efforts in such places could be difficult and life-threatening because of the difficulty in accessing such areas and the dangers presented by unstable structures (NDRRMC, 2023).

Conventional rescue equipment, such as wheeled robots, often fails to operate effectively in constricted, irregular, or blocked paths. Therefore, rescue missions must sometimes resort to taking possible risks to their own safety to come to the aid of trapped victims (Balanay & Ramos, 2024). This problem emphasizes the need to develop new solutions that enable rescuers to operate in restricted areas without risking their own safety (Patalinjug, 2022).

To counter these issues, the authors propose a life-saving rescue robot with a caterpillar-like structure. This invention is based on bio-inspired robotics. The purpose of this invention is to navigate through constricted spaces with a caterpillar-like motion. In this manner, the invention will be able to access spaces where humans and regular robots cannot (Garcia et al., 2023). The invention will be made with efficiency in mind for search and rescue activities (DOST-ASTI, 2024).

The robot will also consist of sensing devices, a microphone, and a speaker that aid in the identification of people trapped and two-way communication. This helps rescue teams gauge the conditions of the victims, reassure them, and develop an effective plan for rescue (Santos & Reyes, 2025). The robot prevents direct human entry into a hazardous environment and thus aids in improving rescue operations (Dela Cruz, 2024).

In disaster-prone countries, like the Philippines, which experiences frequent earthquakes, floods, and structural collapses, a prompt and effective rescue operation is highly needed. Improvement in the accessibility of victims through a caterpillar-like rescue robot provides promise for increasing support to emergency respondents during life-saving operations. Indeed, this may prove robotic technology as a strengthened preparedness and response against disasters.

Research Objectives

The main objective of this study is to design and develop a bio-inspired, caterpillar-like rescue robot capable of navigating hazardous and constricted environments to assist in search and rescue operations.

The specific objectives are enumerated as follows:

1. To achieve a functional caterpillar-like rescue robot capable of navigating narrow, confined, and hazardous spaces such as collapsed buildings, tunnels, and rubble during search and rescue operations;
2. To successfully design and integrate sensing and communication components, specifically a high-definition camera, microphone, and speaker, to effectively locate and establish two-way communication with trapped victims; and:

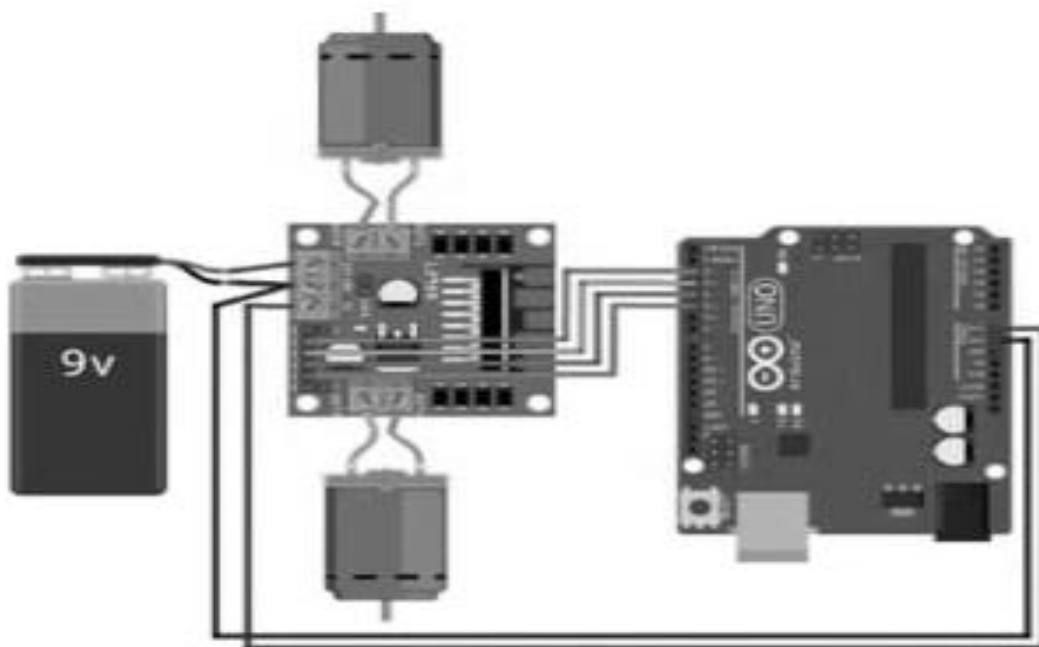
3. To reduce risks to human rescuers by enabling remote exploration of hazardous environments, eliminating the need for direct human entry.

2. Literature Review

Presented in this section are the related literature that consists of inventions made by other researchers that are relevant to the modelling of our own invention.

A. Title: Arduino-Based Control Systems for Multi-Segmented Mobile Robots

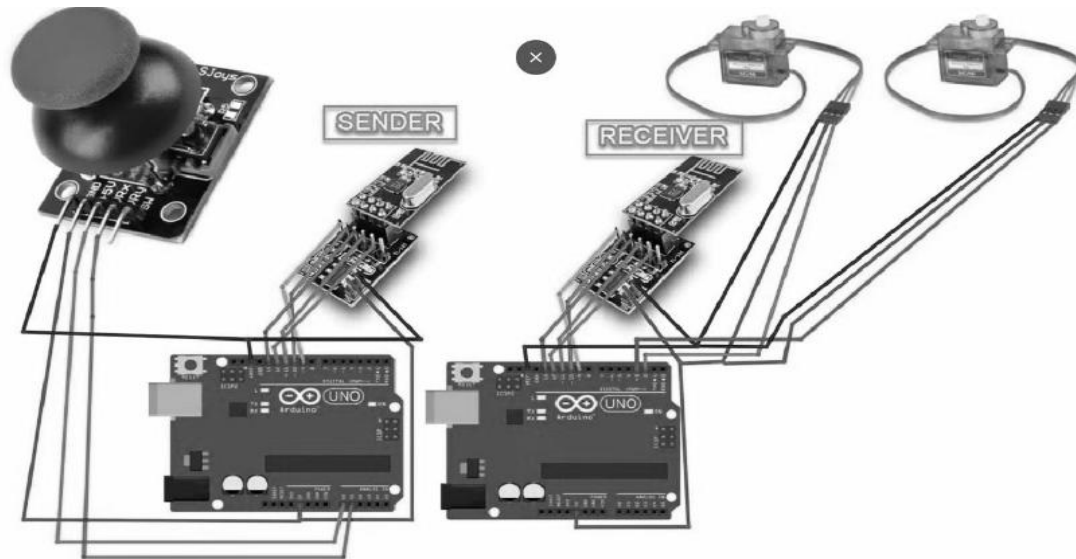
Purpose: To design an integrated control system using microcontrollers and motor drivers that can manage multiple DC motors to simulate biological movement. The 12-volt battery lasts for over 12 hours before needing to be replaced.



Results and Discussion: Similarly, the research by Garcia et al. (2023), using a centralized microcontroller, allows for the synchronization of segments, ensuring the robot can navigate constricted or irregular paths where wheeled robots typically fail.

B. Title: Wireless Remote Operation and Communication in Disaster Zones

Purpose: To establish a reliable two-way communication link and remote-control interface using wireless modules to allow rescuers to operate from a safe distance.



Results and Discussion: Integrating a joystick-controlled transmitter with a receiver on the robot enables real-time steering through debris, providing a remote-operated alternative for exploring dangerous environments. This communication helps rescue teams gauge the conditions of victims, reassure them, and develop an effective plan for rescue. Furthermore, the use of such technology prevents direct human entry into hazardous environments, thereby improving the safety and efficiency of rescue operations (Santos & Reyes, 2020).

C. Title: Antenna Communication System

Purpose: The purpose of the Antenna Communication System is to enable the invention to transmit data wirelessly over greater distances. The antennas increase distance coverage by 100-10,000 meters, enhancing signal transmission and reception, allowing the system to function efficiently without physical connections.



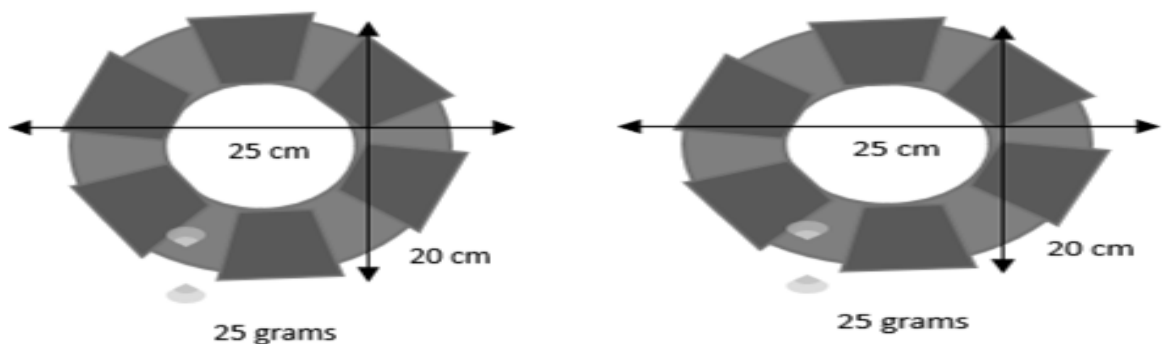
Results and Discussion: The results demonstrate that the antenna effectively improves distance coverage and signal stability. This increased range allows the invention to maintain reliable communication even when components are farther apart, making it suitable for practical wireless applications (Sagouspe & Writer, 2020).

3. Development and Evaluation of the Project

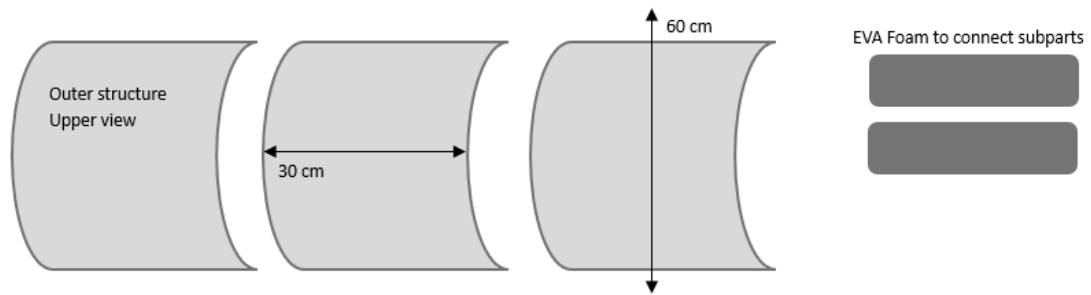
Presented in this section are the developmental procedures that are needed to be undergone in realizing our project, including the design, construction timeframe, costing, and research design.

3.1. Design Phase

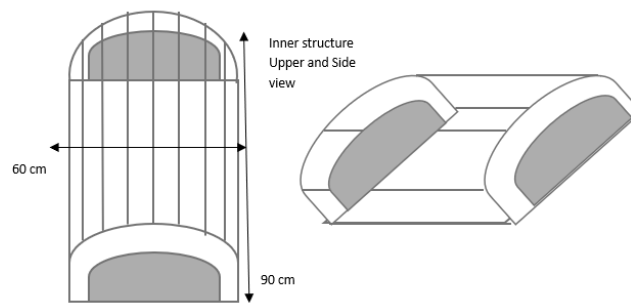
A. Wheel structure



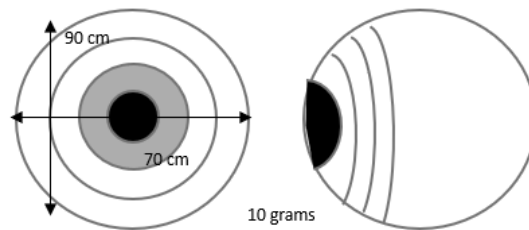
B. Outer structure (Upper View)



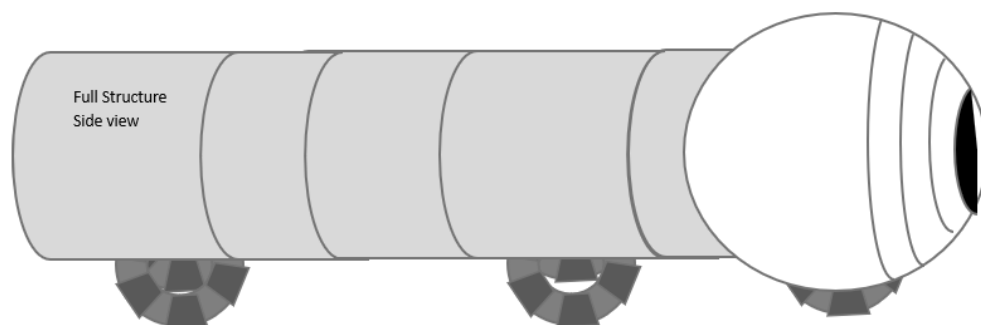
C. Inner structure (upper and side view)



D. Head structure (Front and side view)



E. Full structure (side view)



3.2 Construction Time Frame

	Month 1				Month 2			
	W1	W2	W3	W4	W1	W2	W3	W4
Designing the plan								
Finding parts and components								
Functionality testing								
Tweaks and revision of defects								
Project Evaluation								

	Month 1				Month 2			
	W1	W2	W3	W4	W1	W2	W3	W4
Designing the plan								
Finding parts and components								
Functionality testing								
Tweaks and revision of defects								
Project Evaluation								

3.3 Costing

QUANTITY	UNIT	NAME AND DESCRIPTION	COST (PHP)
6	pcs	common toy car wheels with ridges	420
2	pcs	DC motor standard	410
1	pcs	TB6612FNG dual motor driver	180
2	pcs	servo motor standard	350
1	pcs	joystick module	180
2	pcs	jumper wires	190
1	pcs	NRF24Lo1 Wireless module	220
1	pcs	arduino CH3406	300
1	pcs	12 volts battery	40
1	pcs	small lens camera	800
1	pcs	remote controlled RC	400
TOTAL			3,490

3.4 Research Design

This study adopts the research structure: design and development research (DDR), specifically product and tool research. Design and development research is a type of inquiry unique to the instructional design and technology field dedicated to the creation of new knowledge and the validation of existing practice. This type of research focuses on the products and tools, and research on design and development models (Richey & Klein, 2014). This is compatible with our research because we are developing a product that must be thoroughly and meticulously designed.

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